US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12410732
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ożóg; Mateusz et al.

Flexible support for aircraft motor bearing

Abstract

An assembly is provided for an aircraft motor. The assembly includes a bearing, a stationary structure and a flexible support. The bearing extends circumferentially around an axis. The stationary structure circumscribes the bearing. The flexible support is arranged radially between the bearing and the stationary structure. The flexible support includes an inner ring, an outer ring, a bridge, an open first channel and an open second channel. The inner ring radially engages the bearing. The outer ring radially engages the stationary structure. The bridge projects radially out from the inner ring to the outer ring. The bridge extends circumferentially about the inner ring. The open first channel extends axially into the flexible support from a first side of the flexible support to the bridge. The open second channel extends axially into the flexible support from a second side of the flexible support to the bridge.

Inventors: **Ożóg; Mateusz (Hucisko, PL), Tahvilian; Amir Masoud (Mississauga, CA)**

Applicant: **Pratt & Whitney Canada Corp.** (Longueuil, CA)

Family ID: **94382989**

Assignee: **PRATT & WHITNEY CANADA CORP. (Longueuil, CA)**

Appl. No.: **18/420095**

Filed: **January 23, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250237157 A1	Jul. 24, 2025

Publication Classification

Int. Cl.: **F01D25/16** (20060101)

U.S. Cl.:

CPC **F01D25/164** (20130101); F05D2240/50 (20130101)

Field of Classification Search

CPC: F01D (25/16); F01D (25/162); F01D (25/164); F16C (27/04)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3512856	12/1969	Robinson	384/535	F16C 33/6651
3549220	12/1969	Robinson	384/535	F16C 25/06
3640591	12/1971	Eklund	384/535	F16C 19/163
3709570	12/1972	Galbato	384/536	F16C 27/04
4429923	12/1983	White	384/99	F16C 27/045
4496252	12/1984	Horler	N/A	N/A
4952076	12/1989	Wiley, III	N/A	N/A
5112143	12/1991	Ide	384/107	F16F 15/0237
5421655	12/1994	Ide	N/A	N/A
5455778	12/1994	Ide	700/182	F16J 15/3432
8337090	12/2011	Herborth	N/A	N/A
9926975	12/2017	Smedresman	N/A	N/A
10450893	12/2018	Polly	N/A	N/A
10753391	12/2019	Smedresman	N/A	F16F 15/0237
11021998	12/2020	Ganiger	N/A	N/A
11105223	12/2020	Ganiger	N/A	N/A
11193390	12/2020	Ganiger	N/A	N/A
11674397	12/2022	Ganiger	N/A	N/A
2008/0056633	12/2007	Stout	N/A	N/A
2010/0220948	12/2009	Singh	N/A	N/A
2012/0213629	12/2011	Rouesne	415/170.1	F16C 27/045
2017/0145855	12/2016	Ganiger	N/A	F16C 35/067
2017/0276173	12/2016	Smedresman	N/A	F16F 15/022
2021/0040865	12/2020	Ganiger	N/A	F01D 25/18
2022/0243616	12/2021	Smedresman	N/A	N/A

OTHER PUBLICATIONS

EP search report for EP25153669.4 dated Jul. 2, 2025. cited by applicant

---

Primary Examiner: Elliott; Topaz L.

Attorney, Agent or Firm: Getz BalichLLC

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to an aircraft motor and, more particularly, to a bearing support member for use within the aircraft motor.

BACKGROUND INFORMATION

(2) An aircraft motor such as a gas turbine engine or electric motor may include a flexible bearing support to accommodate slight radial shifts between a rotating structure and a stationary structure of the aircraft motor. Various types and configurations of flexible bearing supports are known in the art. While these known flexible bearing supports have various benefits, there is still room in the art for improvement.

SUMMARY

(3) According to an aspect of the present disclosure, an assembly is provided for an aircraft motor. The assembly includes a bearing, a stationary structure and a flexible support. The bearing extends circumferentially around an axis. The stationary structure circumscribes the bearing. The flexible support is arranged radially between the bearing and the stationary structure. The flexible support includes an inner ring, an outer ring, a bridge, an open first channel and an open second channel. The inner ring radially engages the bearing. The outer ring radially engages the stationary structure. The bridge projects radially out from the inner ring to the outer ring. The bridge extends circumferentially about the inner ring. The open first channel extends axially into the flexible support from a first side of the flexible support to the bridge. The open second channel extends axially into the flexible support from a second side of the flexible support to the bridge.

(4) According to another aspect of the present disclosure, an apparatus is provided for an aircraft motor. The apparatus includes a flexible support for a bearing in the aircraft motor. The flexible support extends axially along an axis between a support first side and a support second side. The flexible support includes an inner ring, an outer ring, a bridge, a first channel and a second channel. The inner ring extends axially along and circumferentially around the axis. The outer ring extends axially along and circumferentially around the axis. A groove projects radially into the outer ring towards the bridge. The groove extends axially within the outer ring with an axial width greater than an axial thickness of the bridge. The groove extends circumferentially around the axis within the outer ring. The bridge extends radially between and is formed integral with the inner ring and the outer ring. The bridge extends circumferentially about the inner ring. The first channel projects axially into the flexible support from the support first side to the bridge. The first channel extends circumferentially around the axis within the flexible support. The second channel projects axially into the flexible support from the support second side to the bridge. The second channel extends circumferentially around the axis within the flexible support.

(5) According to still another aspect of the present disclosure, another apparatus is provided for an aircraft motor. The apparatus includes a flexible support for a bearing in the aircraft motor. The flexible support extends axially along an axis between a support first side and a support second side. The flexible support extends radially between a support inner side and a support outer side. The flexible support includes an inner ring, an outer ring, a bridge, an annular first channel and an annular second channel. The inner ring extends axially along and circumferentially around the axis. The inner ring is disposed at the support inner side. The outer ring extends axially along and circumferentially around the axis. The outer ring is disposed at the support outer side. The bridge projects radially out from the inner ring to the outer ring. The bridge extends circumferentially about the axis. A radial height of the bridge is greater than an axial thickness of the bridge. The axial thickness of the bridge is greater than a radial thickness of the inner ring and/or a radial thickness of the outer ring. The annular first channel projects axially into the flexible support from the support first side to the bridge. The annular second channel projects axially into the flexible support from the support second side to the bridge.

(6) An undercut may project radially into the outer ring from the groove. The undercut may extend

axially within the outer ring. The undercut may extend circumferentially around the axis within the outer ring.
(7) The bearing may be configured as or otherwise include a rolling element bearing with an outer race. The inner ring may circumscribe and radially engage the outer race.
(8) The inner ring may circumscribe and radially contact the bearing.
(9) The assembly may also include a fluid source fluidly coupled with a fluid passage that extends radially through the flexible support. The fluid source may be configured to direct fluid through the fluid passage to provide a fluid buffer radially between the inner ring and the bearing.
(10) The stationary structure may circumscribe and radially contact the outer ring.
(11) A groove may project radially into the outer ring towards the bridge. The groove may extend axially within the outer ring. The groove may extend circumferentially about the axis within the outer ring.
(12) The groove may have an axial width that is greater than an axial thickness of the bridge.
(13) The groove may have a radial depth that is less than a radial height of the bridge.
(14) The flexible support may be attached to the stationary structure through an interference fit at a radial interface between the outer ring and the stationary structure.
(15) The flexible support may be attached to the stationary structure through a bonded connection between the outer ring and the stationary structure.
(16) The flexible support may be attached to the stationary structure with one or more fasteners.
(17) The bridge may circumscribe the inner ring.
(18) The bridge may be a first bridge. The flexible support may also include a second bridge. The second bridge may project radially out from the inner ring to the outer ring. The second bridge may extend circumferentially about the axis. The second bridge may be circumferentially spaced from the first bridge by a gap.
(19) A radial height of the bridge may be greater than an axial thickness of the bridge.
(20) The bridge may be axially aligned with an axial center of the bearing.
(21) The bearing may include a plurality of rolling elements arranged circumferentially about the axis in an array. The bridge may axially overlap the array of the rolling elements.
(22) The outer ring may include a fuse axially spaced from the bridge.
(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a sectional illustration of a portion of an assembly for an aircraft motor.
(2) FIG. **2** is a sectional illustration of a portion of a flexible support.
(3) FIG. **3** is a side view illustration of the flexible support.
(4) FIG. **4** is a sectional illustration of a portion of the flexible support with an offset bridge arrangement.
(5) FIG. **5** is a sectional illustration of a portion of the assembly with another flexible support mounting arrangement.
(6) FIG. **6** is a side view illustration of a portion of the flexible support with multi-bridges.
(7) FIG. **7** is a sectional illustration of a portion of the flexible support of FIG. **6**.
(8) FIG. **8** is a sectional illustration of a portion of the flexible support with a fuse.
(9) FIG. **9** is a sectional illustration of a portion of the assembly with a fluid system.
(10) FIG. **10** is a sectional illustration of a portion of another assembly for the aircraft motor.

DETAILED DESCRIPTION

(11) FIG. **1** illustrates an assembly **20** for an engine of an aircraft. The aircraft may be an airplane, a helicopter, a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. The aircraft motor may be configured as, or otherwise included as part of, a propulsion system for the aircraft. The aircraft motor may also or alternatively be configured as, or otherwise included as part of, an electrical power system for the aircraft. The engine assembly **20** of FIG. **1** includes a stationary structure **22** of the aircraft motor and a rotating structure **24** of the aircraft motor. This engine assembly **20** also includes a mounting assembly **26** for rotatably mounting the rotating structure **24** to the stationary structure **22**. The mounting assembly **26** of FIG. **1** includes a bearing **28** and a flexible support **30** for the bearing **28**; e.g., a flexible bearing support.

(12) The stationary structure **22** may be configured as, or otherwise included as part of, a casing structure of the aircraft motor. The stationary structure **22** of FIG. **1**, for example, includes an engine case **32**, a stationary outer mounting land **34** and a land support **36** (e.g., an annular frame, an array of struts, etc.) extending radially between and structurally tying the outer mounting land **34** to the engine case **32**, where the engine case **32** and the land support **36** are schematically shown. The outer mounting land **34** extends axially along an axis **38**. Briefly, this axis **38** may be a centerline axis of the aircraft motor and/or one or more of its members **22**, **24**, **26**, **28** and/or **30**. The axis **38** may also or alternatively be a rotational axis of the rotating structure **24**. The outer mounting land **34** projects radially inward (towards the axis **38**) to a radial inner side **40** of the outer mounting land **34**. The outer mounting land **34** extends circumferentially about (e.g., completely around) the axis **38**, which provides the outer mounting land **34** with, for example, a full-hoop (e.g., tubular or annular) geometry.

(13) The outer mounting land **34** of FIG. **1** includes an interior land surface **42** disposed at the land inner side **40**. This interior land surface **42** may have a regular cylindrical geometry. The land inner side **40** and its interior land surface **42** partially or completely form a radial outer peripheral boundary of an internal bore **44** in the stationary structure **22** and its outer mounting land **34**. This land bore **44** extends axially along the axis **38** in (e.g., into, within or through) the stationary structure **22** and its outer mounting land **34**.

(14) The rotating structure **24** may be configured as, other otherwise included as part of, a spool or other rotating assembly of the aircraft motor. The rotating structure **24** of FIG. **1**, for example, includes an inner mounting land **46** extending axially in the land bore **44**. This inner mounting land **46** may be part (e.g., an axial section) of an engine shaft. The inner mounting land **46** may alternatively be configured as, or otherwise included as part of, another component mounted onto or formed integral with the engine shaft; e.g., a shaft sleeve, a rotor disk, etc. The inner mounting land **46** extends axially along the axis **38**. The inner mounting land **46** projects radially outward (away from the axis **38**) to a radial outer side **48** of the inner mounting land **46**. The inner mounting land **46** extends circumferentially about (e.g., completely around) the axis **38**, which provides the inner mounting land **46** with, for example, a full-hoop (e.g., tubular or annular) geometry.

(15) The inner mounting land **46** of FIG. **1** includes an exterior land surface **50** disposed at the land outer side **48**. The exterior land surface **50** may have a regular cylindrical geometry.

(16) The bearing **28** may be configured as a rolling element bearing. The bearing **28** of FIG. **1**, for example, includes a bearing inner race **52**, a bearing outer race **54** and a plurality of bearing rolling elements **56** arranged circumferentially about the axis **38** in an array; e.g., a circular array. The inner race **52** extends axially along and circumferentially about (e.g., circumscribes) the inner mounting land **46** and the axis **38**. This inner race **52** is mounted onto (e.g., fixed to) and is rotatable with the rotating structure **24** and its inner mounting land **46**. The outer race **54** is spaced radially outboard from the inner race **52**. The outer race **54** extends axially along and circumferentially about (e.g., circumscribes) the inner race **52**, the array of rolling elements **56** and the axis **38**. The array of rolling elements **56** are arranged radially between the inner race **52** and the outer race **54**, where each of the rolling elements **56** is operable to radially engage and roll

against the inner race **52** and the outer race **54**. The present disclosure, however, is not limited to such an exemplary rolling element bearing configuration. Moreover, while the bearing **28** is described above as the rolling element bearing, it is contemplated the bearing **28** may alternatively be configured as a journal bearing or any other type of bearing suitable for use in the aircraft motor.

(17) Referring to FIG. **2**, the flexible support **30** extends axially along the axis **38** between and to a first side **58** of the flexible support **30** and a second side **60** of the flexible support **30**. The flexible support **30** extends radially from a radial inner side **62** of the flexible support **30** to a radial outer side **64** of the flexible support **30**. The flexible support **30** of FIG. **3** extends circumferentially about (e.g., completely around) the axis **38**, which provides the flexible support **30** with, for example, a full-hoop (e.g., annular) geometry. Referring to FIG. **2**, the flexible support **30** may be configured with an I-beam cross-sectional geometry when viewed, for example, in a reference plane parallel with (e.g., including) the axis **38**. The flexible support **30** of FIG. **2**, for example, includes a radial inner ring **66**, a radial outer ring **68** and an intermediate bridge **70**.

(18) The support inner ring **66** is disposed at the support inner side **62**. The support inner ring **66** of FIG. **2**, for example, extends radially between and to a radial outer side **72** of the support inner ring **66** and the support inner side **62**. The support inner ring **66** extends axially along the axis **38** between and to opposing axial sides **74** and **76** of the support inner ring **66**. The inner ring first side **74** may be disposed at (or axially recessed from) the support first side **58**. The inner ring second side **76** may be disposed at (or axially recessed from) the support second side **60**. The support inner ring **66** of FIG. **3** extends circumferentially about (e.g., completely around) the axis **38**, which provides the support inner ring **66** with, for example, a full-hoop (e.g., tubular) geometry. With this arrangement, the support inner ring **66** of FIG. **2** may form an inner flange of the I-beam cross-sectional geometry.

(19) The support inner ring **66** has an axial width **78** and a radial thickness **80**. The inner ring width **78** is measured axially along the axis **38** from the inner ring first side **74** to the inner ring second side **76**. This inner ring width **78** may be uniform (constant) as the support inner ring **66** extends circumferentially about the axis **38**. The inner ring thickness **80** is measured radially from the support inner side **62** to the inner ring outer side **72**. This inner ring thickness **80** may be uniform as the support inner ring **66** extends circumferentially about the axis **38**.

(20) The flexible support **30** and its support inner ring **66** of FIG. **2** include an interior support surface **82** disposed at the support inner side **62**. The interior support surface **82** may have a regular cylindrical geometry which extends axially from (or near) the inner ring first side **74** to (or near) the inner ring second side **76**.

(21) The support outer ring **68** is disposed at the support outer side **64**. The support outer ring **68** of FIG. **2**, for example, extends radially between and to a radial inner side **84** of the support outer ring **68** and the support outer side **64**. The support outer ring **68** extends axially along the axis **38** between and to opposing axial sides **86** and **88** of the support outer ring **68**. The outer ring first side **86** may be disposed at (or axially recessed from) the support first side **58** and/or axially aligned with the inner ring first side **74**. The outer ring second side **88** may be disposed at (or axially recessed from) the support second side **60** and/or axially aligned with the inner ring second side **76**. The support outer ring **68** of FIG. **3** extends circumferentially about (e.g., completely around) the axis **38**, which provides the support outer ring **68** with, for example, a full-hoop (e.g., tubular) geometry. With this arrangement, the support outer ring **68** of FIG. **2** may form an outer flange of the I-beam cross-sectional geometry.

(22) The support outer ring **68** has an axial width **90** and a radial thickness **92**. The outer ring width **90** is measured axially along the axis **38** from the outer ring first side **86** to the outer ring second side **88**. This outer ring width **90** may be uniform as the support outer ring **68** extends circumferentially about the axis **38**. The outer ring width **90** of FIG. **2** is equal to the inner ring width **78**; however, it is contemplated the outer ring width **90** may alternatively be different (e.g., greater) than the inner ring width **78** (e.g., see FIG. **5**). The outer ring thickness **92** is measured

radially from the support outer side **64** to the outer ring inner side **84**. This outer ring thickness **92** may be uniform as the support outer ring **68** extends circumferentially about the axis **38**. The outer ring thickness **92** of FIG. **2** is equal to the inner ring thickness **80**; however, it is contemplated the outer ring thickness **92** may alternatively be different (e.g., greater or less) than the inner ring thickness **80**.

(23) The support outer ring **68** of FIG. **2** is configured with a groove **94** at the support outer side **64**. The support outer ring **68** of FIG. **2**, for example, includes a base **96** and one or more feet **98** and **100**; e.g., ribs, rims, flanges, etc. The outer ring base **96** extends axially along the axis **38** from the outer ring first side **86** to the outer ring second side **88**. The outer ring base **96** is disposed at the outer ring inner side **84**. The outer ring base **96** of FIG. **2**, for example, extends radially from an outer side **102** of the outer ring base **96** (e.g., a distal inner side of the groove **94**) to the outer ring inner side **84**. The outer ring feet **98** and **100** are disposed at the opposing sides **86** and **88** of the support outer ring **68** and are connected to (e.g., formed integral with) the outer ring base **96**. Each of the outer ring feet **98** and **100** projects radially out from the outer ring base **96** to the support outer side **64**. Each of the outer ring members **96**, **98**, **100** may extend circumferentially about (e.g., completely around) the axis **38**. With this arrangement, the groove **94** projects partially radially into the support outer ring **68** from the support outer side **64** to the outer ring base outer side **102**. The groove **94** extends axially along the axis **38** within the support outer ring **68** between the outer ring feet **98** and **100**. The groove **94** extends circumferentially about (e.g., completely around) the axis **38** within the support outer ring **68**.

(24) The outer ring base **96** has a radial thickness **104** measured radially from the outer ring inner side **84** to the outer ring base outer side **102**. This outer ring base thickness **104** may be uniform as the outer ring base **96** extends circumferentially about the axis **38**. The base thickness **104** of FIG. **2** is different (e.g., less) than the inner ring thickness **80**; however, it is contemplated the outer ring base thickness **104** may alternatively be equal to the inner ring thickness **80**.

(25) The groove **94** has an axial width **106** and a radial depth **108**. The groove width **106** is measured axially along the axis **38** from the first outer ring foot **98** to the second outer ring foot **100**. This groove width **106** may be uniform as the groove **94** extends circumferentially about the axis **38**. The groove width **106** of FIG. **2** is less than (e.g., between 50% and 90% of) the outer ring width **90**. The groove depth **108** is measured radially from the support outer side **64** to the outer ring base outer side **102**. This groove depth **108** may be uniform as the groove **94** extends circumferentially about the axis **38**. The groove depth **108** of FIG. **2** is equal to the outer ring base thickness **104**; however, it is contemplated the groove depth **108** may alternatively be different (e.g., greater or less) than the outer ring base thickness **104**.

(26) The support bridge **70** is disposed radially between and connected to (e.g., formed integral with) the support inner ring **66** and the support outer ring **68** and its outer ring base **96**. The support bridge **70** of FIG. **2**, for example, projects radially out from the support inner ring **66** at its inner ring outer side **72** to the support outer ring **68** at its outer ring inner side **84**. The support bridge **70** extends axially along the axis **38** between and to opposing axial sides **110** and **112** of the support bridge **70**. The bridge first side **110** is axially recessed from each of the first sides **58**, **74**, **86**. The bridge second side **112** is axially recessed from each of the second sides **60**, **76**, **88**. The support bridge **70** of FIG. **2**, for example, is axially centered along the support inner ring **66** and/or the support outer ring **68**; e.g., axially centered between the support sides **58** and **60**. The support bridge **70** of FIG. **3** extends circumferentially about (e.g., completely around) the axis **38**, which provides the support bridge **70** with, for example, a full-hoop (e.g., annular) geometry. With this arrangement, the support bridge **70** of FIG. **2** may form a web of the I-beam cross-sectional geometry.

(27) The support bridge **70** has an axial thickness **114** and a radial height **116**. The bridge thickness **114** is measured axially along the axis **38** from the bridge first side **110** to the bridge second side **112**. This bridge thickness **114** may be uniform as the support bridge **70** extends circumferentially

about the axis **38**. The bridge thickness **114** of FIG. **2** is less than the inner ring width **78**, the outer ring width **90**, the groove width **106** and/or the bridge height **116**. The bridge thickness **114** of FIG. **2**, however, is greater than the inner ring thickness **80**, the outer ring thickness **92**, the outer ring base thickness **104** and/or the groove depth **108**. The bridge height **116** is measured radially from the support inner ring **66** at its inner ring outer side **72** to the support outer ring **68** at its outer ring inner side **84**. This bridge height **116** may be uniform as the support bridge **70** extends circumferentially about the axis **38**. The bridge height **116** of FIG. **2** is greater than the inner ring thickness **80**, the outer ring thickness **92**, the outer ring base thickness **104** and/or the groove depth **108**. The bridge height **116** of FIG. **2**, however, is less than inner ring width **78**, the outer ring width **90** and/or the groove width **106**. The present disclosure, however, is not limited to each of the foregoing exemplary dimensional relationships.

(28) At least (or only) the support inner ring **66**, the support outer ring **68** and the support bridge **70** of FIG. **2** collectively form a first channel **118** in the flexible support **30**. The first channel **118** may be an open channel (e.g., an unfilled channel) in the flexible support **30**. The first channel **118** is disposed at the support first side **58**. The first channel **118** of FIG. **2**, for example, projects axially along the axis **38** into the flexible support **30** from the support first side **58** to the support bridge **70** at its bridge first side **110**. The first channel **118** extends radially within the flexible support **30** from the support inner ring **66** at its inner ring outer side **72** to the support outer ring **68** at its outer ring inner side **84**. The first channel **118** extends circumferentially about (e.g., completely around) the axis **38**, which provides the first channel **118** with, for example, a full-hoop (e.g., annular) geometry.

(29) The first channel **118** has an axial depth **120** measured from the support first side **58** to the bridge first side **110**. This first channel depth **120** may be uniform as the first channel **118** extends circumferentially about the axis **38**. The first channel depth **120** of FIG. **2** is greater than the bridge thickness **114** and/or the bridge height **116**.

(30) At least (or only) the support inner ring **66**, the support outer ring **68** and the support bridge **70** of FIG. **2** collectively form a second channel **122** in the flexible support **30**. The second channel **122** may be an open channel (e.g., an unfilled channel) in the flexible support **30**. The second channel **122** is disposed at the support second side **60**. The second channel **122** of FIG. **2**, for example, projects axially along the axis **38** into the flexible support **30** from the support second side **60** to the support bridge **70** at its bridge second side **112**. The second channel **122** extends radially within the flexible support **30** from the support inner ring **66** at its inner ring outer side **72** to the support outer ring **68** at its outer ring inner side **84**. The second channel **122** extends circumferentially about (e.g., completely around) the axis **38**, which provides the second channel **122** with, for example, a full-hoop (e.g., annular) geometry.

(31) The second channel **122** has an axial depth **124** measured from the support second side **60** to the bridge second side **112**. This second channel depth **124** may be uniform as the second channel **122** extends circumferentially about the axis **38**. The second channel depth **124** of FIG. **2** is greater than the bridge thickness **114** and/or the bridge height **116**. The second channel depth **124** of FIG. **2** is equal to the first channel depth **120**; however, it is contemplated the second channel depth **124** may alternatively be different (e.g., greater or less) than the first channel depth **120**; e.g., see FIG. **4** where the support bridge **70** is arranged off-center along the support inner ring **66** and/or the support outer ring **68**.

(32) Referring to FIG. **1**, the flexible support **30** mounts (e.g., fixedly attaches) the bearing **28** and its outer race **54** to the stationary structure **22** and its outer mounting land **34**. The support inner ring **66** of FIG. **1**, for example, extends axially along (e.g., axially overlaps) and circumferentially around (e.g., circumscribes) the bearing **28** and its outer race **54**. The support inner ring **66** and its surface **82** further radially engage (e.g., radially contact, abut radially against, etc.) the bearing **28** and its outer race **54**. Here, the outer race **54** is attached to the support inner ring **66** by an interference fit at a radial interface between the outer race **54** and the support inner ring **66** at the

support inner side **62**. Similarly, the outer mounting land **34** extends axially along and circumferentially around the flexible support **30** and its support outer ring **68**. The support outer ring **68** and its outer ring feet **98** and **100** further radially engage (e.g., radially contact, abut radially against, etc.) the stationary structure **22** and its outer mounting land **34**. Here, the support outer ring **68** is attached to the outer mounting land **34** by an interference fit at a radial interface between the support outer ring **68** at its support outer side **64** and the outer mounting land **34** and its land inner side **40**.

(33) With the foregoing arrangement, the flexible support **30** is configured to accommodate (e.g., slight) radial movement between the rotating structure **24** and the stationary structure **22** during aircraft motor operation. The outer ring base **96** of FIG. **1**, for example, may operate as a flexible member (e.g., a spring diagram, etc.) connecting the support bridge **70** to the outer ring feet **98** and **100**. To facilitate a radial upward shift of the rotating structure **24** in FIG. **1**, the outer ring base **96** may deform and bow radially outward into the groove **94**/the groove depth **108** of FIG. **2** may locally decrease. To facilitate a radial downward shift of the rotating structure **24** in FIG. **1**, the outer ring base **96** may deform and bow radially inward away from the groove **94**/the groove depth **108** of FIG. **2** may locally increase. Of course, opposite deformation of the outer ring base **96** may occur at a diametrically opposing side of the flexible support **30** to the section shown in FIG. **1**. Dimensions of the flexible support **30** may be selected to tune a spring rate of the outer ring base **96**.

(34) The support bridge **70** of FIG. **1** is axially aligned with a center of the bearing **28**. The support bridge **70**, however, may alternatively be axially offset from the center of the bearing **28**. Even with such offset arrangements, however, the support bridge **70** may axially overlap the array of rolling elements **56**.

(35) While the flexible support **30** is described above as being attached to the bearing **28** and the stationary structure **22** through interference fits, the present disclosure is not limited to such exemplary mounting techniques. For example, the flexible support **30** and its support outer ring **68** may also or alternatively be bonded (e.g., welded, brazed, etc.) to the stationary structure **22** and its outer mounting land **34**. In another example, referring to FIG. **5**, the flexible support **30** may also or alternatively be mechanically fastened to the stationary structure **22** and its outer mounting land **34** by one or more mechanical fasteners **126**; e.g., bolts. A mounting flange **128**, for example, may be connected to and project radially out from the support outer ring **68** at the support first side **58**/the outer ring first side **86**. This mounting flange **128** may abut axially against or otherwise axially engage the stationary structure **22** and its outer mounting land **34**. The mounting flange **128** is attached to the stationary structure **22** and its outer mounting land **34** by the mechanical fasteners **126**.

(36) In some embodiments, referring to FIG. **3**, the support bridge **70** may be configured as an annular element of the flexible support **30**. In other embodiments, referring to FIG. **6**, the support bridge **70** may be one of multiple support bridges **70** which extend radially between and connect the support inner ring **66** and the support outer ring **68**. Here, each of the support bridges **70** extends partially circumferentially about the axis **38** configuring each support bridge **70** as an arcuate element of the flexible support **30**. These support bridges **70** of FIG. **6** are arranged (e.g., an equispaced) circumferentially about the axis **38** in an array and form a plurality of ports **130**; e.g., air gaps. Each port **130** of FIG. **6** is formed by and extends laterally (e.g., circumferentially) between a respective circumferentially neighboring (e.g., adjacent) pair of the support bridges **70**. Referring to FIG. **7**, each port **130** extends axially along the axis **38** between and to the first channel **118** and the second channel **122**; e.g., axially through the array of support bridges **70**.

(37) In some embodiments, referring to FIG. **8**, the flexible support **30** may be configured with a fuse **132**; e.g., a fracture feature. The support outer ring **68** of FIG. **8**, for example, includes an undercut **134** (e.g., a notch, a groove, a channel, etc.) in the outer ring base **96**. This undercut **134** projects partially radially into the outer ring base **96** from the groove **94** and the outer ring base

outer side **102**. The undercut **134** extends axially within the outer ring base **96**. The undercut **134** extends circumferentially about (e.g., completely around) the axis **38** within the outer ring base **96**. The undercut **134** may thereby form a stress point (e.g., a weak point) in the support outer ring **68** and its outer ring base **96**. With this configuration, the support outer ring **68** may fracture when subject to relatively high (e.g., extreme) loading. This fracturing may facilitate further control radial movement (e.g., shifting) between the rotating structure **24** and the stationary structure **22**. Moreover, the frangibility in the flexible support **30** may also cut a load path between the stationary structure **22** and rotating structure **24** (see FIG. **1**). This may reduce vibration transmitted by the aircraft motor to the airframe in case of an engine event resulting in an unbalanced rotor; e.g., a blade off event. Note, while the fuse **132** is shown on a top-righthand side of the support bridges **70** in FIG. **8**, it is contemplated the fuse **132** and its undercut **134** (see dashed lines) may alternatively (or also) be on a top-lefthand side, a bottom-righthand side and/or a bottom-lefthand side of the support bridges **70**.
(38) In some embodiments, referring to FIG. **9**, the flexible support **30** may include one or more fluid passages **136** (e.g., lubricant passages) arranged circumferentially about the axis **38** in an array; e.g., a circular array. Each of these fluid passages **136** may project radially through the flexible support **30**. Each fluid passage **136**, for example, may extend (e.g., sequentially) radially through the support members **68**, **70** and **68** from the groove **94** to the support inner side **62**. The fluid passages **136** may be fluidly coupled with a fluid source **138**; e.g., a lubricant source. With this arrangement, each fluid passage **136** may direct fluid (e.g., lubricant) received from the fluid source **138** into a gap **140** formed by and radially between the support inner ring **66** and the outer race **54**. The fluid in the gap **140** may form a fluid buffer (e.g., a film, a cushion) radially between the support inner ring **66** and the outer race **54**. The flexible support **30** and its support inner ring **66** may thereby be connected to and support the bearing **28** and its outer race **54** through a fluid damper.
(39) In some embodiments, referring to FIG. **1**, the flexible support **30** may be configured as an insert radially between and mounting the bearing **28** and its outer race **54** to the stationary structure **22** and its outer mounting land **34**. The flexible support **30** of the present disclosure, however, is not limited to such an exemplary flexible support configuration. For example, referring to FIG. **10**, the flexible support **30** may alternatively be configured as a bump stop support for another flexible mount **142** for the bearing **28**; e.g., a squirrel cage bearing mount. Here, the flexible support **30** and the flexible mount **142** are independently attached to the stationary structure **22**. Moreover, the flexible support **30** is spaced radially outboard from, axially overlaps and circumscribes the flexible mount **142**. With this arrangement, the flexible support **30** provides a progressive bump stop for the flexible mount **142** under certain conditions.
(40) The aircraft motor of the present disclosure may be configured as or otherwise include a gas turbine engine. The gas turbine engine may be a geared gas turbine engine with a geartrain operatively coupling one or more shafts to one or more rotors in a fan section, a compressor section and/or any other engine section. Alternatively, the gas turbine engine may be configured without a geartrain as a direct-drive gas turbine engine. The gas turbine engine may include a single spool or multiple spools. The gas turbine engine may be configured as a turbofan engine, a turbojet engine, a turboprop engine, a turboshaft engine, a propfan engine, a pusher fan engine or any other type of gas turbine engine. The gas turbine engine may alternatively be configured as an auxiliary power unit (APU). The present disclosure therefore is not limited to any particular types or configurations of gas turbine engines. Moreover, while the aircraft motor is described above as a gas turbine engine, it is contemplated the aircraft motor may alternatively be configured as or otherwise include another type of internal combustion (IC) engine such as, but not limited to, a rotary engine (e.g., a Wankel type engine) or a reciprocating piston engine, or even an electric motor.
(41) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible

within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An assembly for an aircraft motor, comprising: a bearing extending circumferentially around an axis; a stationary structure circumscribing the bearing; and a flexible support arranged radially between the bearing and the stationary structure, the flexible support including an inner ring radially engaging an outer surface of the bearing; an outer ring radially engaging the stationary structure; a bridge projecting radially out from the inner ring to the outer ring, the bridge extending circumferentially about the inner ring; an open first channel extending axially into the flexible support from a first side of the flexible support to the bridge; and an open second channel extending axially into the flexible support from a second side of the flexible support to the bridge; wherein a continuous annular groove projects radially into the outer ring towards the bridge, the groove extends axially within the outer ring, and the groove extends circumferentially about the axis within the outer ring; and wherein the groove has an axial width that is greater than an axial thickness of the bridge.
2. The assembly of claim 1, wherein the bearing comprises a rolling element bearing with an outer race; and the inner ring circumscribes and radially engages the outer race.
3. The assembly of claim 1, wherein the inner ring circumscribes and radially contacts the bearing.
4. The assembly of claim 1, further comprising: a fluid source fluidly coupled with a fluid passage that extends radially through the flexible support; the fluid source configured to direct fluid through the fluid passage to provide a fluid buffer radially between the inner ring and the bearing.
5. The assembly of claim 1, wherein the stationary structure circumscribes and radially contacts the outer ring.
6. The assembly of claim 1, wherein the groove has a radial depth that is less than a radial height of the bridge.
7. The assembly of claim 1, wherein the flexible support is attached to the stationary structure through an interference fit at a radial interface between the outer ring and the stationary structure.
8. The assembly of claim 1, wherein the flexible support is attached to the stationary structure through a bonded connection between the outer ring and the stationary structure.
9. The assembly of claim 1, wherein the flexible support is attached to the stationary structure with one or more fasteners.
10. The assembly of claim 1, wherein the bridge circumscribes the inner ring.
11. The assembly of claim 1, wherein the bridge is a first bridge, and the flexible support further includes a second bridge; and the second bridge projects radially out from the inner ring to the outer ring, the second bridge extends circumferentially about the axis, and the second bridge is circumferentially spaced from the first bridge by a gap.
12. The assembly of claim 1, wherein a radial height of the bridge is greater than an axial thickness of the bridge.
13. The assembly of claim 1, wherein the bridge is axially aligned with an axial center of the bearing.
14. The assembly of claim 1, wherein the bearing comprises a plurality of rolling elements arranged circumferentially about the axis in an array; and the bridge axially overlaps the array of the plurality of rolling elements.
15. The assembly of claim 1, wherein the outer ring comprises a fuse axially spaced from the

bridge.

16. The assembly of claim 1, wherein the bridge has an axial thickness that is greater than at least one of a radial thickness of the inner ring or a radial thickness of the outer ring.

17. An assembly for an aircraft motor, comprising: a bearing extending circumferentially around an axis; a stationary structure circumscribing the bearing; and a flexible support arranged radially between the bearing and the stationary structure, the flexible support including an inner ring with an interior support surface radially engaging the bearing; an outer ring radially engaging the stationary structure; a bridge projecting radially out from the inner ring to the outer ring, the bridge extending circumferentially about the inner ring; an open first channel extending axially into the flexible support from a first side of the flexible support to the bridge; and an open second channel extending axially into the flexible support from a second side of the flexible support to the bridge; wherein a continuous annular groove projects radially into the outer ring towards the bridge, the groove extends axially within the outer ring, and the groove extends circumferentially about the axis within the outer ring; wherein the groove has an axial width that is greater than an axial thickness of the bridge; and wherein the bridge has an axial thickness that is greater than at least one of a radial thickness of the inner ring or a radial thickness of the outer ring.